IET Signal Processing

Research Article

Bayesian step least mean squares algorithm for Gaussian signals

ISSN 1751-9675
Received on 10th February 2020
Revised 8th June 2020
Accepted on 29th June 2020
E-First on 4th September 2020
doi: 10.1049/iet-spr.2020.0058
www.ietdl.org

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Abstract: Selecting the step of the least mean squares (LMS) algorithm is an old problem. This study uses a new approach to address this problem resulting in a new algorithm with excellent system identification performance. The LMS algorithm, with time-varying step, size can be shown to be equivalent to the Kalman filter in some conditions. This is as long as the state noise of the Kalman filter and the step size of the LMS algorithm are chosen carefully. The Kalman filter is the optimum linear estimator (Bayesian) given the state and the measurement noise covariance matrices, but these matrices are not always known. This work considers the case where these matrices are not known, in the special cases that the Kalman filter reduces to the LMS. This results in an algorithm to select the step-size of the LMS algorithm with few priors. The optimum step size can be calculated using estimates of the probability density function (PDF) of the coefficient estimation error variance (q_w) and measurement noise variance (q_v). The PDFs can be estimated from the data using Bayes' rule and assuming Gaussian reference and measurement noise signals. The resulting algorithm to determine q_w and q_v is a second small Kalman filter, and the outputs of this filter (means and covariances) are used to determine the expected value of the step.

1 Introduction

Selecting the step size of the least mean squares (LMS) [1] algorithm is an old problem, with many papers published on the subject [2–7]. However, recent results show that the LMS is equivalent to the Kalman filter (KF) if the state noise of the KF and the step of the LMS algorithm are carefully selected [5]. This results in a step size that is a function of the current average coefficient estimation variance q_w and measurement error variance q_v. In this work, the probability density functions (PDFs) of these variables are estimated given the data and used to calculate the expected value of the step, resulting in an optimal value for the step given the measurements and the approximations used. Source code is available in [8].

Regarding notation, vectors and matrices are shown in bold–italic font as in $\boldsymbol{x}$ and bold–italic font uppercase as in $\boldsymbol{A}$, respectively. $\boldsymbol{x}^{\mathrm{T}}$ represents the transpose of vector $\boldsymbol{x}$. The expected values of a random process $x(n)$ is expressed as $\mathrm{E}[x(n)]$, and the time average by $\langle x(n)\rangle$. The absolute value of a real number x is $|x|$. The identity matrix is represented by $\boldsymbol{I}$.

This work is based on a connection between the NLMS and the KF published in [9] that allows choosing the step size of the NLMS. Several other works [7, 10–12] have made similar connections and reached similar algorithms. In [7], they optimise the step to minimise the system misalignment, resulting in the JO-LMS algorithm. They also estimate the state noise using the adaptive filter coefficients changes. In [11] the expectation-maximisation (EM) [13–15] algorithm is used. The expectation step of the EM corresponds to the KF. Approximation of this step, as described by the authors, gives the same results as in [9]. Then the authors use the maximisation step of the EM to obtain formulas for the measurement and state noise. In [12] they analyse the algorithm as presented in [10]. Furthermore, there are many papers related to these. In [16], it proposed to use just two values of the step instead of continually changing it, and the results are compared with the JO-LMS. Only the case when the reference is white is considered. The possibility of adaptive select the regularisation parameter of the NLMS instead of the step size discussed in [17]. The measurement noise is taken to be known. An Adaptation of these algorithms for bilinear forms is described in [18], along with a comparison with the KF for the same problem. Finally, Sulthana *et al*. [19] described an application of the KLMS algorithm and reduced complexity forms in cardiac signal analysis.

Several works have derived algorithms to estimate the state and/or the measurement noise used in the KF. In some of these works [20, 21] all the measurements' data are stored and processed offline iteratively using the EM algorithm. However, this requires significant amounts of computations, storage, and delay. Other methods use online estimation. These can also be divided into parallel multi-model methods and adaptive methods (ADs). In multi-model methods [22–26] a set of KFs with different sets of parameters are run in parallel, and the likelihood of each model is evaluated using the measures. The resulting estimate is a weighted average of the estimates using each set of parameters. ADs use the information on innovations, measures, or state estimates to estimate the noise covariance matrices.

For instance, in [27], the autocorrelation function of the innovations' process is used to obtain the noise covariance matrices. However, the method has a high dependence on assuming that the measurement noise is white, which is not always the case. Also, it considers only the filter in steady-state with constant measurement and state transition matrix. In [28, 29] they derive formulas for the covariance matrix of the measures $\boldsymbol{C}$ and invert them to obtain the state noise covariance matrix $\boldsymbol{Q}$ assuming the measurement noise matrix $\boldsymbol{R}$ is known. In [25] $\boldsymbol{Q}$ and $\boldsymbol{R}$ that maximise the likelihood of each measurement are obtained assuming the other is known ($\boldsymbol{R}$ or $\boldsymbol{Q}$). This results in functions of the innovation sequence for the calculation of $\boldsymbol{R}$ and of the state estimates for the calculation of $\boldsymbol{Q}$. All these papers assume that $\boldsymbol{R}$ is known to estimate $\boldsymbol{Q}$ or (in [27]) that the measurement noise is white, mostly because distinguishing state and measurement noises is a hard problem.

Other works have derived formulas for the step-size of the NLMS using different approaches. For instance, in [30] the step size is chosen proportional to the residual noise power and in [31] to the correlation of the residual noise $e(n)$ with its delay version $e(n-1)$. For large filter, the optimal step for the NLMS can be approximated by $\mu = (q_e - q_v)/q_e$ [6] where q_e is the residual noise

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signal power and q_v is the measurement noise signal power, and this simple formula motives some practical formulas for the step. Mader *et al*. [2] proposed to use

$$\mu(n) = \frac{\beta(n)\langle u^2(n)\rangle}{\boldsymbol{u}^{\mathrm{T}}(n)\boldsymbol{u}(n)\langle e^2(n)\rangle} \tag{1}$$

with $\beta(n) = \langle\langle e^2(n)\rangle/\langle u^2(n)\rangle\rangle$. This will be referred to as Mader algorithm. Carini and Malatini [6] proposed (in the active noise control (ANC) context, and also based in [2]) selecting the step using

$$\mu(n) = \frac{\hat{n}(n)}{\boldsymbol{u}^{\mathrm{T}}(n)\boldsymbol{u}(n)\langle e^2(n)\rangle} \tag{2}$$

with $\boldsymbol{m}(n) = \langle e(n)\boldsymbol{u}(n)/p_{\mathrm{t}}(n)\rangle$, $\hat{n}(n) = \langle e(n)\boldsymbol{m}^{\mathrm{T}}(n)\boldsymbol{u}(n)\rangle$. This will be referred to as Carini algorithm.

In both algorithms, the time average is calculated using first-order IIR filters with large forgetting factors λ. Also related is the work from [3] that uses different approximations for the same optimal step. The work from Benesty *et al*. [4] used a different optimality criterion: equalling the residual error power to the measuring noise power. This criterion results in different but related expressions.

2 Algorithm derivation

The LMS algorithm [1] is described by

$$\boldsymbol{w}(n) = \boldsymbol{w}(n-1) + \mu\, e(n)\boldsymbol{u}(n) \tag{3}$$

where $e(n) = d(n) - \boldsymbol{u}^{\mathrm{T}}(n)\boldsymbol{w}(n-1)$ is the error signal, $d(n)$ is the desired signal, $\boldsymbol{u}(n) = [u(n)\ldots u(n-N+1)]^{\mathrm{T}}$ is the reference signal vector, $u(n)$ is the reference signal $\boldsymbol{w}(n) = [w_0(n)\ldots w_{N-1}(n)]^{\mathrm{T}}$ is the vector with the adaptive filter coefficients, μ is the step size and N is the adaptive filter size. The step size can be made time varying, and following the work in [5, 7] the step size should be equal to

$$\mu(n) = \frac{1}{p_t(n) + q_v(n)/q_w(n)} \tag{4}$$

where $p_{\mathrm{t}}(n) = \boldsymbol{u}^{\mathrm{T}}(n)\boldsymbol{u}(n)$, q_v is the measurement noise variance, q_w is the variance of the current estimate of each coefficient. As discussed in [5], using this step, $\boldsymbol{w}(n)$ is the optimum minimum mean square error (MMSQ) estimate of the filter for a carefully chosen state noise:

$$\boldsymbol{w}(n) = \mathrm{E}[\boldsymbol{x}(n)|q_w(n), q_v(n), \boldsymbol{w}(n-1)], \tag{5}$$

where $\boldsymbol{x}(n) = \boldsymbol{w}_{\mathrm{o}}(n)$ is the vector with the optimal values for the filter coefficients. This is since the LMS is, in fact, equivalent to a KF. If, however, $q_w(n)$ and $q_v(n)$ are unknown and are only estimated from the data, then the optimum MMSQ estimate of the filter coefficients $\boldsymbol{w}_b(n)$, assuming $\boldsymbol{w}(n-1)$ is known, is

$$\begin{aligned}\boldsymbol{w}_b(n) &= \mathrm{E}[\boldsymbol{x}(n)|\boldsymbol{w}(n-1)]\\ &= \int\int^{\mathrm{E}}[\boldsymbol{x}(n)|q_w(n), q_v(n), \boldsymbol{w}(n-1)]\\ &\quad \rho_q(q_w(n), q_v(n), n)\mathrm{d}q_w(n)\mathrm{d}q_v(n)\end{aligned} \tag{6}$$

where $\rho_q(q_w(n), q_v(n), n)$ is the joint PDF of $q_w(n)$ and $q_v(n)$ given the data up to and including time n. Then

$$\boldsymbol{w}_b(n) = \int\int^{\boldsymbol{w}}(n)\rho_q(q_w(n), q_v(n), n)\mathrm{d}q_w(n)\mathrm{d}q_v(n) \tag{7}$$

and

$$\boldsymbol{w}_b(n) = \boldsymbol{w}_b(n-1) + \mu_b(n)\, e(n)\boldsymbol{u}(n) \tag{8}$$

with

$$\mu_b(n) = \mathrm{E}[\mu(n)] = \int\int \mu(n)\rho_q(q_w(n), q_v(n), n)\mathrm{d}q_w(n)\mathrm{d}q_v(n) \tag{9}$$

since only $\mu(n)$ depends on $q_w(n)$ or $q_v(n)$ in (3). Note that this means that some care should be taken when calculating the step μ_b. The step should be calculated using (9), and not, for instance, using the estimates of q_v and q_w directly in (4) although this can lead to acceptable results in most cases. Also, $\mu(n)$ could be calculated using Monte Carlo methods. Namely, samples could be taken from the distribution of $q_w(n)$ and $q_v(n)$, μ calculated using (4) and the mean would be calculated at the end. However, in this work, μ is calculated directly by integrating (9).

2.1 State variables PDF

In order to calculate $\mu_b(n)$, $\rho_q(q_w(n), q_v(n), n)$ needs to be known. In the following, an expression for it is obtained, assuming Gaussian signals. Let $\boldsymbol{q}(n) = [q_w(n), q_v(n)]^{\mathrm{T}}$ and $\boldsymbol{a}(n) = [p_{\mathrm{t}}(n), 1]^{\mathrm{T}}$, then since, in the chosen model, all the filter coefficients have an equal mean square error $q_w(n)$, $\mathrm{E}[e^2(n)] = \boldsymbol{a}^{\mathrm{T}}(n)\boldsymbol{q}(n)$. Since, $e(n)$ is zero-mean Gaussian, and using Bayes' rule, results in

$$\begin{aligned}\rho_{\boldsymbol{q}(n)|n}(\boldsymbol{q}) &= \rho_{\boldsymbol{q}(n)|e(n),e(n-1),\ldots}(\boldsymbol{q})\\ &= \frac{\rho_{e(n)|\boldsymbol{q}(n)}(\boldsymbol{q})\rho_{\boldsymbol{q}(n)|n-1}(\boldsymbol{q})}{\rho_{e(n)|n-1}(\boldsymbol{q})} = k\frac{\mathrm{e}^{-(e^2(n)/2\boldsymbol{a}^{\mathrm{T}}(n)\boldsymbol{q})}}{\sqrt{2\pi\,\boldsymbol{a}^{\mathrm{T}}(n)\boldsymbol{q}}}\rho_{\boldsymbol{q}(n)|n-1}(\boldsymbol{q})\end{aligned} \tag{10}$$

where $\rho_x(y)$ denotes the PDF of the random variable x at the point y, and $\rho_{x|n}(y)$ denotes the PDF of x given measurements up to time n as described. Assuming, $\rho_{\boldsymbol{q}(n)|n-1}(\boldsymbol{q}) = \rho_{\boldsymbol{q}(n-1)|n-1}(\boldsymbol{q})$ that corresponds to consider that $\boldsymbol{q}(n)$ does not change with time [It is known that $\boldsymbol{q}(n)$ does change in time in LMS but this approximation is only used to get a candidate for the PDF of $\boldsymbol{q}(n)$], (10) can also be written as

$$\rho_{\boldsymbol{q}|n}(\boldsymbol{q}) = k\left(\prod_{i=1}^{n}\frac{\mathrm{e}^{-(e^2(i)/2\boldsymbol{a}^{\mathrm{T}}(i)\boldsymbol{q})}}{\left((2\pi)^2\boldsymbol{q}^{\mathrm{T}}\boldsymbol{a}(i)\boldsymbol{a}^{\mathrm{T}}(i)\boldsymbol{q}\right)^{1/4}}\right)\rho_{\boldsymbol{q}|0}(\boldsymbol{q})\,. \tag{11}$$

That suggest approximating $\rho_{\boldsymbol{q}(n)|n}(\boldsymbol{q})$ by a PDF of the form

$$\rho_{\boldsymbol{q}(n)|n}(\boldsymbol{q}) \simeq \hat{\rho}_{\boldsymbol{q}}(\boldsymbol{q}, \boldsymbol{D}(n), \boldsymbol{c}(n), n) = k_0\frac{\mathrm{e}^{-(n/2\boldsymbol{c}^{\mathrm{T}}(n)\boldsymbol{q})}}{\left((2\pi)^2\boldsymbol{q}^{\mathrm{T}}\boldsymbol{D}(n)\boldsymbol{q}\right)^{n/4}}\,. \tag{12}$$

where k_0 is a normalisation constant that does not depend on $\boldsymbol{q}$, $\boldsymbol{c}(n) = [c_1(n), c_2(n)]^{\mathrm{T}}$ and

$$\boldsymbol{D}(n) = \begin{pmatrix} d_{1,1}(n) & d_{1,2}(n)\\ d_{2,1}(n) & d_{2,2}(n)\end{pmatrix} \tag{13}$$

This was, in part, selected because it allows to obtain a closed-form expression for the PDF of μ, $\rho_\mu(\mu)$. Using (4), (12) and the formula for distribution of two functions of two random variables [32] $\rho_\mu(z)$ is given by

$$\begin{aligned}\rho_\mu(z) &= \int\frac{\hat{\rho}_{\boldsymbol{q}}(\boldsymbol{q}, \boldsymbol{D}, \boldsymbol{c}, n)}{z^2/q_w}\mathrm{d}q_w\\ &= \frac{1}{4}\pi^{-n/2}\Gamma\left(\frac{n}{2}-2\right)\left(\frac{-c_2zp_t + c_1z + c_2}{n}\right)^{(n-4)/2}\\ &\quad \left(d_{2,2}(zp_{\mathrm{t}}-1)^2 + z(zd_{1,1} - 2d_{1,2}(zp_{\mathrm{t}}-1))\right)^{-n/4}\end{aligned} \tag{14}$$

for $n > 4$, where $\Gamma(n)$ is the Gamma function. For large $n(n-4 \simeq n)$, the value of z that maximises $\rho_\mu(z)$, z_0, is given by (4)

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as expected. However, the authors could not calculate $\mathrm{E}[\rho_\mu(z)]$ in closed form for arbitrary n. So, to obtain an approximate result for $\mathrm{E}[\rho_\mu(z)]$, $\rho_\mu(z)$ was approximated by a Gaussian function $\hat{\rho}_\mu(z)$ around z_0, using the first and second derivatives of $\log(\rho_\mu(z))$: $\delta_1(z)$, $\delta_2(z)$. The expressions for these derivatives are easily derived but are rather long so they are not shown here. Equalling the first and second derivatives of $\log(\hat{\rho}_\mu(z))$ and $\log(\rho_\mu(z))$ results that $\hat{\rho}_\mu(z)$ has mean $z_1 = z_0 - \delta_1(z_0)/\delta_2(z_0)$ and variance $q_z = -1/\delta_2(z_0)$. Let $p_z = 1/q_z$. The expected value can be calculated by integrating

$$\mathrm{E}[\mu] \simeq \int_0^{1/p_t} z\hat{\rho}_\mu(z)\mathrm{d}z \tag{15}$$

resulting in

$$a = \frac{\sqrt{p_z}(1 - z_1 p_t)}{\sqrt{2}p_t}, \quad b = \frac{z_1\sqrt{p_z}}{\sqrt{2}}, \tag{16}$$

$$\mathrm{E}[\mu] \simeq z_1 + \frac{-2\mathrm{e}^{-a^2} + 2\mathrm{e}^{-b^2}}{\sqrt{2\pi p_z}(\mathrm{erf}(a) + \mathrm{erf}(b))} \tag{17}$$

where $\mathrm{erf}(x)$ is the Gauss error function.

2.2 Measure update equations

It remains to determine how to update $\boldsymbol{D}$ and $\boldsymbol{c}$. To do this, $\hat{\rho}_{\boldsymbol{q}}(\boldsymbol{q}, \boldsymbol{D}(n), \boldsymbol{c}(n), n)$ is also approximated by a Gaussian. Note that the estimator that results from this approximation is the KF. Since, the KF is the best linear estimator, even for non-Gaussian PDFs, the resulting estimator has the same property. Let

$$\hat{\rho}_{\boldsymbol{q}}(\boldsymbol{q}, \boldsymbol{D}, \boldsymbol{c}, n) \simeq \tilde{\rho}_{\boldsymbol{q}}(\boldsymbol{q}, \boldsymbol{A}, \boldsymbol{b}) = k_1 \mathrm{e}^{\boldsymbol{b}^{\mathrm{T}}\boldsymbol{q} - \frac{1}{2}\boldsymbol{q}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{q}} \tag{18}$$

where k_1 a normalisation constant $\boldsymbol{A}^{-1}$ is the multivariate Gaussian covariance matrix and $\boldsymbol{q}_o = \boldsymbol{A}^{-1}\boldsymbol{b}$ is the mean.

To make the approximation accurate, $\boldsymbol{A}$ and $\boldsymbol{b}$ are selected, so that: (i) The values of $\boldsymbol{q}$ that maximises both functions are equal. Note that for $\tilde{\rho}_{\boldsymbol{q}}$ this is the mean $\boldsymbol{q}_o$. (ii) The Hessian matrices of the logarithm of both functions $(\ln \hat{\rho}_{\boldsymbol{q}} \text{ and } \ln \tilde{\rho}_{\boldsymbol{q}})$ are equal at the maximum point $\boldsymbol{q}_o$. Note that, the Hessian matrix of the logarithm of the Gaussian distribution is $-\boldsymbol{A}$. The gradient of $\ln \hat{\rho}_{\boldsymbol{q}}$ is

$$\nabla \ln(\hat{\rho}_{\boldsymbol{q}}(\boldsymbol{q})) = -\frac{n}{2}\left(-\frac{\boldsymbol{c}}{(\boldsymbol{c}^{\mathrm{T}}\boldsymbol{q})^2} + \frac{\boldsymbol{D}\boldsymbol{q}}{\boldsymbol{q}^{\mathrm{T}}\boldsymbol{D}\boldsymbol{q}}\right), \tag{19}$$

and the Hessian of $\ln \hat{\rho}_{\boldsymbol{q}}$ is

$$\mathrm{H}\ln(\hat{\rho}_{\boldsymbol{q}}(\boldsymbol{q})) = -\frac{n}{2}\left(\frac{2\boldsymbol{c}\boldsymbol{c}^{\mathrm{T}}}{(\boldsymbol{c}^{\mathrm{T}}\boldsymbol{q})^3} + \frac{\boldsymbol{D}(\boldsymbol{q}^{\mathrm{T}}\boldsymbol{D}\boldsymbol{q}) - 2\boldsymbol{D}\boldsymbol{q}\boldsymbol{q}^{\mathrm{T}}\boldsymbol{D}}{(\boldsymbol{q}^{\mathrm{T}}\boldsymbol{D}\boldsymbol{q})^2}\right). \tag{20}$$

Making (19) equal to zero results in $\boldsymbol{q} = \boldsymbol{D}^{-1}\boldsymbol{c}/(\boldsymbol{c}^{\mathrm{T}}\boldsymbol{D}^{-1}\boldsymbol{c})$, so (1) is equal to

$$\boldsymbol{q}_{\mathrm{o}} = \frac{\boldsymbol{D}^{-1}\boldsymbol{c}}{\boldsymbol{c}^{\mathrm{T}}\boldsymbol{D}^{-1}\boldsymbol{c}} = \boldsymbol{A}^{-1}\boldsymbol{b}. \tag{21}$$

At the maximum, $\mathrm{H}\ln(\hat{\rho}_{\boldsymbol{q}}(\boldsymbol{q}))$ simplifies to (not for instance that $\boldsymbol{c}^{\mathrm{T}}\boldsymbol{q} = 1$ for $\boldsymbol{q} = \boldsymbol{q}_o$)

$$\mathrm{H}\ln(\hat{\rho}_{\boldsymbol{q}}(\boldsymbol{q}_o)) = -\frac{n}{2}\boldsymbol{D}(\boldsymbol{c}^{\mathrm{T}}\boldsymbol{D}^{-1}\boldsymbol{c}) \tag{22}$$

and (2) results in

$$\frac{n}{2}\boldsymbol{D}(\boldsymbol{c}^{\mathrm{T}}\boldsymbol{D}^{-1}\boldsymbol{c}) = \boldsymbol{A}. \tag{23}$$

These equations are underdetermined. One solution is:

$$\boldsymbol{D} = \boldsymbol{A}, \ \xi = \boldsymbol{b}^{\mathrm{T}}\boldsymbol{A}^{-1}\boldsymbol{b} \tag{24}$$

$$\boldsymbol{c} = \frac{\boldsymbol{b}}{\xi}, \ n = 2\xi. \tag{25}$$

This solution will be considered in the text ahead. Other equivalent formulas for ξ are

$$\xi = \boldsymbol{b}^{\mathrm{T}}\boldsymbol{q}_o = \boldsymbol{q}_o^{\mathrm{T}}\boldsymbol{A}\boldsymbol{q}_o \tag{26}$$

Also, let

$$\zeta = \boldsymbol{a}^{\mathrm{T}}\boldsymbol{q}_o. \tag{27}$$

$\zeta(n)$ is an estimate of $\mathrm{E}[e^2(n)]$.

Given a Gaussian approximation for $\hat{\rho}_{\boldsymbol{q}}$, $\tilde{\rho}_{\boldsymbol{q}}$, the PDF of the measure $(e^2(n))$, is also approximated by a Gaussian with mean $\boldsymbol{a}^{\mathrm{T}}(n)\boldsymbol{q}(n)$ and variance $E[e^4(n)] - E[e^2(n)]^2 \simeq 2\zeta^2(n)$. So, (10) becomes

$$\rho_{\boldsymbol{q}(n)|n}(\boldsymbol{q}) \simeq k\frac{\mathrm{e}^{-\left((e^2(n) - \boldsymbol{a}^{\mathrm{T}}(n)\boldsymbol{q}(n))^2/(4\zeta^2(n))\right)}}{\sqrt{4\pi\zeta^2(n)}}\tilde{\rho}_{\boldsymbol{q}}(\boldsymbol{q}). \tag{28}$$

And, the measure update equations become

$$\boldsymbol{A}(n+1) = \boldsymbol{A}(n) + \frac{\boldsymbol{a}(n)\boldsymbol{a}^{\mathrm{T}}(n)}{2\zeta^2(n)} \tag{29}$$

$$\boldsymbol{b}(n+1) = \boldsymbol{b}(n) + \frac{\boldsymbol{a}(n)e^2(n)}{2\zeta^2(n)}. \tag{30}$$

Note that, other non-Gaussian distribution can also be considered by changing $2\zeta^2(n)$ to the appropriate value of the variance of $e^2(n)$, like $k_{\mathrm{D}\zeta^2(n)}$.

2.3 Time update equations

The paper proceeds to discuss the time update equations (Tables 1 and 2). At each time step, the error in the filter coefficient estimates decreases. This decrease can be modelled by [1]

$$\boldsymbol{K}(n+1) = (\boldsymbol{I} - \mu\boldsymbol{R})\boldsymbol{K}(n)(\boldsymbol{I} - \mu\boldsymbol{R}) + \mu^2\boldsymbol{R}q_v \tag{57}$$

where $\boldsymbol{K}(n) = E[(\boldsymbol{w}(n) - \boldsymbol{w}_o(n))^{\mathrm{T}}(\boldsymbol{w}(n) - \boldsymbol{w}_o(n))]$ is the covariance matrix of the filter coefficients estimation error, and $\boldsymbol{w}_o(n)$ is the optimal filter, the vector with the filter coefficients that minimise the error signal variance. Note that this is not the corresponding equation of the KF or KLMS since the last assumes optimal values for the step size.

For small step size, (ignoring μ^2 terms) the equation converges to steady-state value $\boldsymbol{K}(\infty) = q_w(\infty)\boldsymbol{I}$, independent of $\boldsymbol{R}$, and given by $q_w(\infty) = \mu q_v/2$. In this work, this decrease is modelled by

$$q_w(n+1) = (1 - \alpha\mu p(n))^2 q_w(n) + \alpha\mu^2 p(n) q_v(n) \tag{58}$$

where $p(n) = p_t(n)/N$ and $p(n)$ is approximately equal to the average of the eigenvalues of $\boldsymbol{R}$. Constant α compensates for the eigenvalue spread that results in slow converging modes. α reduces the predicted overall convergence speed. Note that, for small steps, the steady-state value of (61) and (62) is the same. For now, the value of $q_v(n)$ is taken to be constant: $q_v(n+1) = q_v(n)$. These equations are joint in a single vector equation $\boldsymbol{q}(n+1) = \boldsymbol{C}(n)\boldsymbol{q}(n)$, with

$$\boldsymbol{C}(n) = \begin{bmatrix} \gamma_1(n)(1 - \alpha\mu(n)p(n))^2 & \alpha\gamma_2(n)\mu^2(n)p(n) \\ 0 & \gamma_2(n) \end{bmatrix} \tag{59}$$

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where $\gamma_1(n)$ and $\gamma_2(n)$ are additional variables that usually have a value of 1, and are further discussed ahead.

The time update equations become

$$\boldsymbol{A}(n+1) = (\boldsymbol{C}^{-1}(n))^{\mathrm{T}}\boldsymbol{A}(n)\boldsymbol{C}^{-1}(n) \tag{60}$$

$$\boldsymbol{b}(n+1) = (\boldsymbol{C}^{-1}(n))^{\mathrm{T}}\boldsymbol{b}(n) \tag{61}$$

and in $\boldsymbol{q}_o(n+1) = \boldsymbol{C}(n)\boldsymbol{q}_o(n)$.

In the estimation process, $q_w(n)$ and $q_v(n)$ can become negative due to fluctuations caused by noise. To prevent this, another technique is used. When $q_w(n)$ or $q_v(n)$ take negative values, their values are changed back to positive by making (in (63)), $\gamma_1(n) = -1$ or $\gamma_2(n) = -1$, respectively, and updating $\boldsymbol{A}(n)$ and $\boldsymbol{b}(n)$ accordingly. This changes the sign of the coefficient and is better than making it zero because, for instance, making q_w zero also makes the step size zero, according to (4), and stalls convergence.

Note that, until now, changes in $q_v(n)$ and in $\boldsymbol{w}_o(n)$ were not considered. State noise usually models these changes (for instance, in the KF), but this work follows a different approach. In the KF, the state noise adds to the state estimation covariance matrix $(\boldsymbol{A}^{-1})$, resulting in a lower bond on $\boldsymbol{A}^{-1}$ and an upper bound on $\boldsymbol{A}$. So, instead trace($\boldsymbol{A}$) is directly limited using $\boldsymbol{A}(n) = k_t\boldsymbol{A}(n)$ and $\boldsymbol{b}(n) = k_t\boldsymbol{b}(n)$ with

$$k_t(n) = \begin{cases} 1 & \text{if } \mathrm{trace}(\boldsymbol{A}(n)) \le T \\ \dfrac{T}{\mathrm{trace}(\boldsymbol{A}(n))} & \text{if } \mathrm{trace}(\boldsymbol{A}(n)) > T \end{cases}. \tag{62}$$

Note that $\boldsymbol{b}(n)$ is also changed so that the value of $\boldsymbol{q}_o(n)$ remains the same.

The resulting algorithm is presented in Table 1. Equation (49) requires some more comments. The expression in (14) is only valid for $n > 4$ so for $n_x \le 4$, $z(n)$ is used as an estimate for the step. Note that n_x is also an estimate of n, so it quickly becomes greater than 4. When $\delta_2(n) > 0$, it makes no sense to make a Gaussian approximation (the second derivative of the log of a Gaussian is constant and negative), so, instead, half of the maximum step is used. Finally, (17) has a high computational cost, and may need to be approximated in an actual implementation, although this has not been tested. Table 2 describes how (17) is calculated in the current implementation. Note that the dependence on n was removed from all variables to simplify notation. The first expression for $\delta_1(z)$ and $\delta_2(z)$ are presented, the next expressions for all the variables used in (17). Finally, even in a double-precision implementation, the calculation of (17) requires the use of alternative expressions, for different ranges of values, to avoid numerical issues as shown. In (74) $\mathrm{erf}(x)$ is replaced by $1 - \mathrm{erfc}(x)$ to prevent problems when

Table 1 Proposed algorithm

Initialisation

$$\boldsymbol{w}(0) = [1, 0, \ldots, 0]^{\mathrm{T}} \tag{31}$$

$$\boldsymbol{q}_o(0) = [1, 1]^{\mathrm{T}} \tag{32}$$

$$\boldsymbol{A} = 0,\ \boldsymbol{b} = 0 \tag{33}$$

$$\delta = \text{small number} \tag{34}$$

Iterate

$$e(n) = d(n) - \boldsymbol{u}(n)^{\mathrm{T}}\boldsymbol{w}(n) \tag{35}$$

$$p_t(n) = \boldsymbol{u}(n)^{\mathrm{T}}\boldsymbol{u}(n),\ p(n) = p_t(n)/N \tag{36}$$

$$\boldsymbol{a}(n) = [p_t(n), 1]^{\mathrm{T}} \tag{37}$$

$$\zeta(n) = \boldsymbol{a}^{\mathrm{T}}(n)\boldsymbol{q}_o(n) \tag{38}$$

$$\boldsymbol{A}(n) = \boldsymbol{A}(n-1) + \frac{\boldsymbol{a}(n)\boldsymbol{a}^{\mathrm{T}}(n)}{2\zeta^2(n)} \tag{39}$$

$$\boldsymbol{b}(n) = \boldsymbol{b}(n-1) + \frac{\boldsymbol{a}(n)e^2(n)}{2\zeta^2(n)} \tag{40}$$

$$\boldsymbol{q}_o(n) = (\boldsymbol{A}(n) + \delta\boldsymbol{I})^{-1}\boldsymbol{b}(n) \tag{41}$$

$$\xi(n) = \boldsymbol{b}(n)^{\mathrm{T}}\boldsymbol{q}_o(n) \tag{42}$$

$$\boldsymbol{c}(n) = \boldsymbol{b}(n)/\xi(n) \tag{43}$$

$$n_{\mathrm{x}}(n) = 2\xi(n) \tag{44}$$

$$[q_w(n), q_v(n)]^{\mathrm{T}} = \boldsymbol{q}_o(n) \tag{45}$$

$$z(n) = \min\left(\max\left(\frac{q_w(n)}{q_w(n)p_t(n) + q_v(n)}, 0\right), \frac{1}{p_t}\right) \tag{46}$$

$$\mu(n) = \begin{cases} z(n) & n_{\mathrm{x}} \le 4 \\ 1/(2p_t) & n_{\mathrm{x}} > 4 \text{ and } \delta_2(n) > 0 \\ \text{calculate (17)} & \text{otherwise} \end{cases} \tag{47}$$

$$\boldsymbol{w}(n+1) = \boldsymbol{w}(n) + \mu(n)\boldsymbol{u}(n)e(n) \tag{48}$$

$$\gamma_1(n) = 1 \quad \text{if} \quad q_w(n) > 0 \quad \text{else} \quad -1 \tag{49}$$

$$\gamma_2(n) = 1 \quad \text{if} \quad q_v(n) > 0 \quad \text{else} \quad -1 \tag{50}$$

$$\boldsymbol{C}(n) = \begin{bmatrix} \gamma_1(n)(1 - \alpha\mu(n)p(n))^2 & \alpha\gamma_2(n)\mu^2(n)p(n) \\ 0 & \gamma_2(n) \end{bmatrix} \tag{51}$$

$$\hat{\boldsymbol{A}}(n+1) = (\boldsymbol{C}(n)^{-1})^{\mathrm{T}}\boldsymbol{A}(n)\boldsymbol{C}(n)^{-1} \tag{52}$$

$$\hat{\boldsymbol{b}}(n+1) = (\boldsymbol{C}(n)^{-1})^{\mathrm{T}}\boldsymbol{b}(n) \tag{53}$$

$$k_t(n) = 1 \text{ if } \mathrm{trace}\left(\hat{\boldsymbol{A}}(n+1)\right) \le T \text{ else } \frac{T}{\mathrm{trace}\left(\hat{\boldsymbol{A}}(n+1)\right)} \tag{54}$$

$$\boldsymbol{A}(n+1) = k_t\hat{\boldsymbol{A}}(n+1) \tag{55}$$

$$\boldsymbol{b}(n+1) = k_t\hat{\boldsymbol{b}}(n+1) \tag{56}$$

Table 2 Proposed Algorithm subroutine to calculate (17)

$$\delta_a = n(-2a_{1,2}(zp_t - 1) + z(a_{1,1} - 2a_{1,2}p_t) + 2a_{2,2}p_t(zp_t - 1) + za_{1,1}) \tag{63}$$

$$\delta_b = 4\left(a_{2,2}(zp_t - 1)^2 + z(za_{1,1} - 2a_{1,2}(zp_t - 1))\right)$$

$$\delta_1(z) = \frac{(n-4)(c_1 - c_2p_t)}{2(-c_2zp_t + c_1z + c_2)} - \frac{\delta_a}{\delta_b}$$

$$\delta_c = n(-2a_{1,2}(zp_t - 1) + z(a_{1,1} - 2a_{1,2}p_t) + 2a_{2,2}p_t(zp_t - 1) + za_{1,1})^2 \tag{64}$$

$$\delta_d = 4\left(a_{2,2}(zp_t - 1)^2 + z(za_{1,1} - 2a_{1,2}(zp_t - 1))\right)^2$$

$$\delta_2(z) = -\frac{n\left(2a_{2,2}p_t^2 - 4a_{1,2}p_t + 2a_{1,1}\right)}{4\left(a_{2,2}(zp_t - 1)^2 + z(za_{1,1} - 2a_{1,2}(zp_t - 1))\right)} + \frac{\delta_c}{\delta_d} - \frac{(n-4)(c_1 - c_2p_t)^2}{2(-c_2zp_t + c_1z + c_2)^2}$$

$$z_1 = z - \delta_1(z)/\delta_2(z) \tag{65}$$

$$p_z = -\delta_2(z) \tag{66}$$

$$a = \frac{\sqrt{p_z}(1 - z_1p_t)}{\sqrt{2}p_t},\ b = \frac{z_1\sqrt{p_z}}{\sqrt{2}}, \tag{67}$$

$$q_w(n)$$

$$\text{if } a + b < \delta \tag{68}$$

$$0$$

$$\text{elseif } |a| > 10 \wedge |b| > 10 \wedge \mathrm{sign}(a)\langle 0 \wedge \mathrm{sign}(b)\rangle 0$$

$$z_1 + \frac{2 - 2\mathrm{e}^{-b^2 + a^2}}{\sqrt{2\pi p_z}\left(\dfrac{\mathrm{e}^{-1/(2a^2)}}{\sqrt{\pi}a} + \dfrac{\mathrm{e}^{-b^2 + a^2 - 1/(2b^2)}}{\sqrt{\pi}b}\right)}$$

$$\text{else if } |a| > 10 \wedge |b| > 10 \wedge \mathrm{sign}(a) > 0 \wedge \mathrm{sign}(b) < 0 \tag{69}$$

$$z_1 - \frac{2 - 2\mathrm{e}^{-a^2 + b^2}}{\sqrt{2\pi p_z}\left(\dfrac{\mathrm{e}^{-1/(2b^2)}}{\sqrt{\pi}b} + \dfrac{\mathrm{e}^{-a^2 + b^2 - 1/(2a^2)}}{\sqrt{\pi}a}\right)}$$

$$\text{elseif } |a| > 3 \wedge |b| > 3 \wedge \mathrm{sign}(a)\mathrm{sign}(b) < 0 \tag{70}$$

$$z_1 + \frac{2\mathrm{e}^{-a^2} - 2\mathrm{e}^{-b^2}}{\sqrt{2\pi p_z}\{\mathrm{sign}(a)\mathrm{erfc}(|a|) + \mathrm{sign}(b)\mathrm{erfc}(|b|)\}}$$

$$\text{else}$$

$$z_1 + \frac{-2\mathrm{e}^{-a^2} + 2\mathrm{e}^{-b^2}}{\sqrt{2\pi p_z}\{\mathrm{erf}(a) + \mathrm{erf}(b)\}} \tag{71}$$

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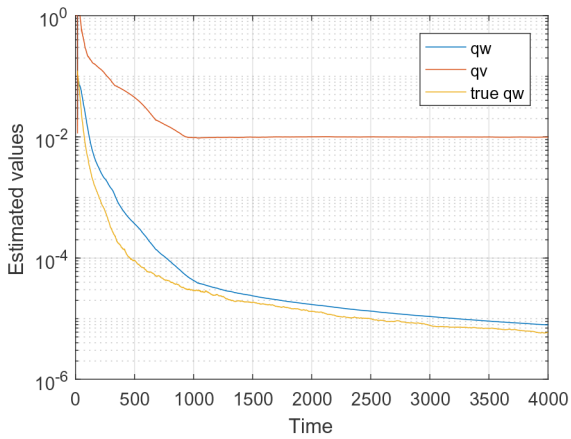

Fig. 1 *Estimation of q_v and q_w. Average of 50 simulations*

Table 3 General simulation parameters

M	100	number of Monte Carlo runs
N	16	filter size
q_v	0.01	measurement noise variance
q_u	1.00	reference signal variance
s_w	1.00	coefficient standard deviation
$\boldsymbol{h}$	$[1, 0.5]$	reference filter impulse response

Table 4 General algorithm parameters

α	0.33	eigenvalue spread compensation factor
T	10^{11}	maximum trace of $\boldsymbol{A}$

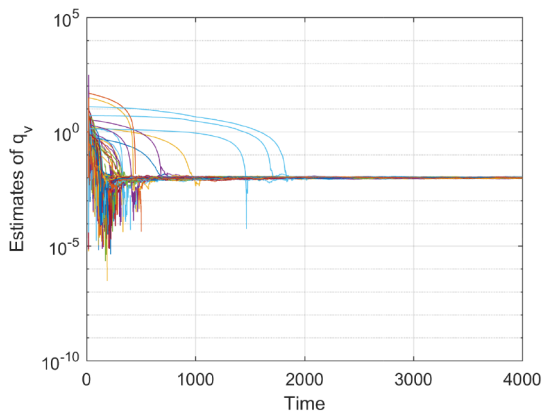

Fig. 2 *Estimates of q_v. Results of 100 single Monte Carlo runs*

$\mathrm{erf}(a)$ and $\mathrm{erf}(b)$ saturate at $+1$ or -1 and subtractive cancellation makes $\mathrm{erf}(a) + \mathrm{erf}(b)$ round to zero. In (72) and (73) $\mathrm{erfc}(x)$ is replaced by $\mathrm{e}^{-x^2 - 1/(2x^2)}/(\sqrt{\pi}x)$ that is a good approximation for $x \gg 1$ and numerator and denominator are multiplied by e^{a^2} or e^{b^2} to prevent problems when even $\mathrm{erfc}(x)$ rounds to zero.

3 Simulation results

This section presents the simulation results. Fig. 1 plots simulation results using the values in Table 3 for the system setup. The error signal is calculated by $e(n) = \boldsymbol{u}^{\mathrm{T}}(n)\boldsymbol{w}_o - \boldsymbol{u}^{\mathrm{T}}(n)\boldsymbol{w}(n) + v(n)$ where the optimal filter has size N and coefficients with sample standard deviation s_w, the measurement signal $v(n)$, has a variance q_v, the reference signal $u(n)$ has variance q_u and is the result of filtering white noise by a filter with impulse response given by $\boldsymbol{h}$. The size of $\boldsymbol{w}(n)$ is also N. The parameters of the algorithms are presented in Table 4. These parameters were optimised by trial and error. It can be seen that q_v and q_w are accurately estimated, and the filter

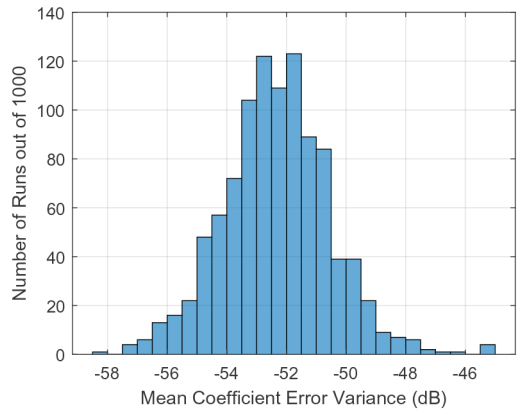

Fig. 3 *Histogram of the final value for the mean coefficient error variance*

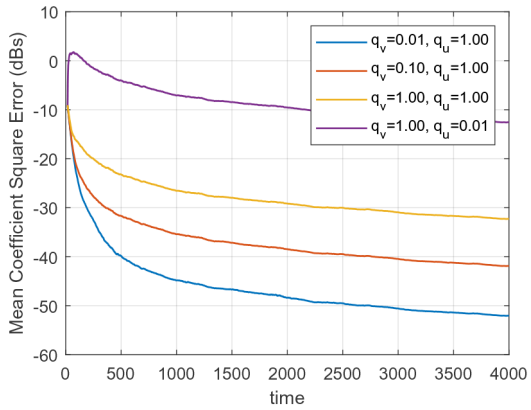

Fig. 4 *Mean coefficient estimation error for different values of true-q_v and q_u*

coefficient estimates mean square error (true-q_w) decreases to very low values.

Fig. 2 shows the time evolution of $q_v(n)$ in 100 simulations with the same parameters as Fig. 1. It can be seen that, in a few cases, $q_v(n)$ maintains a high value for a long time before it converges to the correct value, degrading the average shown in Fig. 1. This happens when an initial low estimate of $q_w(n)/q_v(n)$ causes a very small step size that stalls convergence. More, precisely since the estimated inverse covariance matrix $\boldsymbol{A}$ is initialised to zero, the initial estimates of $q_w(n)$ and $q_v(n)$ are based on the first few measures of $e^2(n)$, and this may result in a low value for $q_w(n)/q_v(n)$. This will result in a low value for the step-size.

This low value will make the $\boldsymbol{C}(n)$ matrix close to identity and the $\boldsymbol{A}(n)$ matrix close to singular, resulting in there being insufficient information to determine q_w and q_v accurately in a short time. This will perpetuate the initial erroneous guesses until even the small amount of information available (that comes from the reference signal power $p_t(n)$ fluctuations) is enough the improve the estimates and increase the step size. This can easily be solved by adding priors information to the estimates of q_w and q_v. For instance, by initialising $\boldsymbol{q}$ to $\boldsymbol{q} = [1, 0.1]^{\mathrm{T}}$, $\boldsymbol{A}$ to $\boldsymbol{A} = \mathrm{diag}([1, 100])$ and $\boldsymbol{b}$ to $\boldsymbol{b} = \boldsymbol{A}\boldsymbol{q}$ the above problem is solved. However, even without prior information, the convergence stalling is not frequent, as seen in Fig. 2. This can also be seen using Fig. 3 that shows the histogram for the final value for true-q_w of 1000 Monte Carlo runs. These values have a low dispersion around -52 dB.

Fig. 4 plots the mean coefficient error variance true-q_w for different values of the true measurement error variance true-q_v and reference signal variance q_u. As expected, true-q_w increases with q_v and decreases with q_u. The algorithm always has good performance since, as shown in Fig. 5, the values of q_v is always correctly estimated after some time.

Figs. 6 and 7 show a comparison with other well-known algorithms for selecting the step size of the LMS algorithm. The

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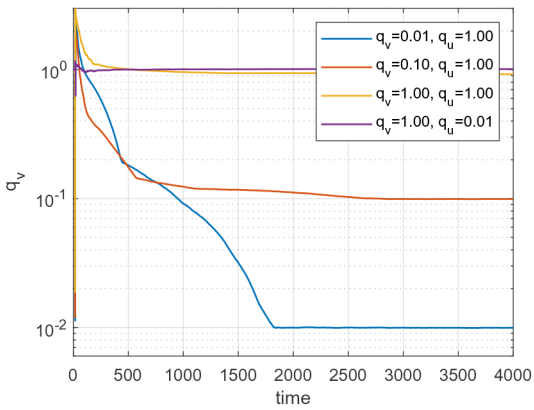

Fig. 5 *Estimate of the measurement noise variance q_v for different values of true- q_v and q_u*

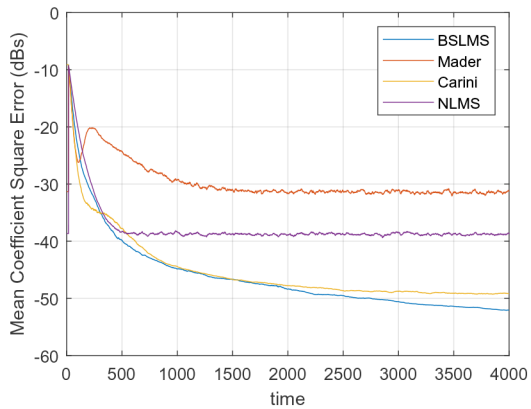

Fig. 6 *Comparison with other algorithms with optimised parameters*

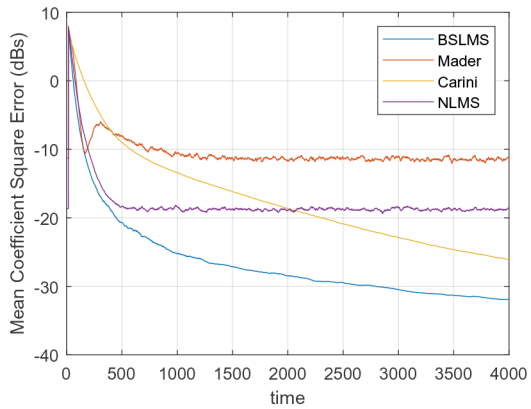

Fig. 7 *Comparison with other algorithms with non-optimised parameters*

parameters of all algorithms were optimised for the values in Table 3. This results in the plot of Fig. 6. In Fig. 7, the following setup parameters were changed: $s_w = 10.0$ and $q_v = 1.0$, while algorithm parameters were kept constant. In both plots, the proposed algorithm shows the best results, although the difference is much bigger in the latter one. The reference colouring filter is the same in both simulations.

4 Conclusion

This presents an algorithm with excellent performance in system identification, namely, with a very small estimation error, thanks to the carefully selected step size of the LMS algorithm. The step size is selected using the expected value of the step of the KLMS algorithm by integrating the corresponding PDF. The expected value of the step can be calculated using the measures and a KF implementation that results from Gaussian approximations of the non-Gaussian PDF of the chosen variables. To obtain estimates of coefficients error variance q_w and measurement noise variance q_v in a short time, it is required to model the decreases in q_w with the convergence of the filter. This work models this decrease, and incorporates it into the KF. Although the algorithm is rather complex, its computational complexity is not prohibitive for long FIR filters.

5 Acknowledgments

This work was supported by national funds through Fundação para a Ciência e a Tecnologia (FCT) with reference UIDB/50021/2020.

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